

Lecture 6: Linear Regression of Time-Series Data III

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Linear Regression of Time-Series Data III

Introduction

Recap: Asymptotic Variance

- We have derived an asymptotic distribution of $\hat{\beta}$.
- Under TS1'–TS3',

$$\sqrt{T}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}(0, AVar(\hat{\beta})),$$

where $AVar(\hat{\beta}) := (E[x_t x_t'])^{-1} \Omega (E[x_t x_t'])^{-1}$ with $\Omega := \lim_{T \rightarrow \infty} \frac{1}{T} Var(\sum_{t=1}^T x_t \varepsilon_t)$.

- In general,

$$\begin{aligned}\Omega &= \lim_{T \rightarrow \infty} \frac{1}{T} \left\{ \sum_{t=1}^T Var(x_t \varepsilon_t) + \sum_{t=1}^T \sum_{s=1}^T Cov(x_t \varepsilon_t, x_s \varepsilon_s) \right\} \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \left\{ \sum_{t=1}^T E[\varepsilon_t^2 x_t x_t'] + \sum_{t=1}^T \sum_{s=1}^T E[\varepsilon_t \varepsilon_s x_t x_s'] \right\}.\end{aligned}$$

- $AVar(\hat{\beta})$ hinges on the assumption about the variance and autocovariance of $\{\varepsilon_t\}$.

Recap: Asymptotic Variance

- To simplify the derivation, we imposed two assumptions:
 - **Homoskedasticity** (TS4'-1): $E[\varepsilon_t^2 | x_t] = \sigma^2$
 - **No serial correlation** (TS4'-2): $E[\varepsilon_t \varepsilon_s | x_t, x_s] = 0$
- These assumptions may not be plausible in time-series analysis.
Ex) an ARMA(1,1) model:

$$\begin{aligned}y_t &= \alpha + \beta y_{t-1} + \gamma e_{t-1} + e_t \\ &= \alpha + \beta y_{t-1} + \varepsilon_t,\end{aligned}$$

where $\varepsilon_t := \gamma e_{t-1} + e_t$ with $e_s \sim WN(0, \sigma_e^2)$ for all $s \leq t$.

$$\implies E[\varepsilon_t \varepsilon_{t-1}] = \gamma E[e_{t-1}^2] = \gamma \sigma_e^2 > 0.$$

$\hookrightarrow$ TS4'-2 is violated...

Outline

- Consequences of Heteroskedasticity and Serial Correlation
- Heteroskedasticity-Autocorrelation-Robust (HAC) Estimator
- Generalized Least Squares (GLS) Estimation
- HAC vs. GLS
- Two Caveats

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Consequences of Heteroskedasticity and Serial Correlation

Setup

- There are four possible configurations:

	Homoskedasticity	No Serial Correlation	$AVar(\hat{\beta})$
Scenario 1	✓	✓	$\sigma^2(E[x_t x_t'])^{-1}$
Scenario 2	✗	✓	?
Scenario 3	✗	✗	?
Scenario 4	✓	✗	?

- Scenario 4 is less likely to be the case in practice.
- We want to study $AVar(\hat{\beta})$ under Scenarios 2 and 3.
- Q. What does Ω look like?

Asymptotic Variance of $\hat{\beta}$: General Expression

- After some algebra, we have

$$\Omega = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \sum_{s=1}^T E[\varepsilon_t \varepsilon_s x_t x_s'] = \dots = \sum_{h=-\infty}^{\infty} \lambda(h),$$

where $\lambda(h) := E[\varepsilon_t \varepsilon_{t-h} x_t x_{t-h}'] = \text{Cov}(\varepsilon_t x_t, \varepsilon_{t-h} x_{t-h})$.

- $\lambda(0)$: Homoskedasticity vs. Heteroskedasticity
- $\lambda(h)$: No serial correlation vs. Serial correlation

► Derivation

Asymptotic Variance of $\hat{\beta}$: General Expression

- **Scenario 1:** Homoskedasticity and No serial correlation

$$\begin{aligned}\Omega &= \lambda(0) + \underbrace{\lambda(-1)}_0 + \underbrace{\lambda(1)}_0 + \underbrace{\lambda(-2)}_0 + \underbrace{\lambda(2)}_0 + \dots \\ &= E[\varepsilon_t^2 x_t x_t'] = E[\underbrace{E[\varepsilon_t^2 | x_t]}_{\sigma^2} x_t x_t'] \\ &= \sigma^2 E[x_t x_t'],\end{aligned}$$

so that

$$AVar(\hat{\beta}) = \sigma^2 (E[x_t x_t'])^{-1}.$$

Asymptotic Variance of $\hat{\beta}$: General Expression

- **Scenario 2:** Heteroskedasticity and No serial correlation

$$\begin{aligned}\Omega &= \lambda(0) + \underbrace{\lambda(-1)}_0 + \underbrace{\lambda(1)}_0 + \underbrace{\lambda(-2)}_0 + \underbrace{\lambda(2)}_0 + \dots \\ &= E[\varepsilon_t^2 x_t x_t'],\end{aligned}$$

so that

$$AVar(\hat{\beta}) = (E[x_t x_t'])^{-1} E[\varepsilon_t^2 x_t x_t'] (E[x_t x_t'])^{-1}.$$

Asymptotic Variance of $\hat{\beta}$: General Expression

- **Scenario 3:** Heteroskedasticity and Serial correlation

$$\begin{aligned}\Omega &= \lambda(0) + \underbrace{\lambda(-1)}_{\neq 0} + \underbrace{\lambda(1)}_{\neq 0} + \underbrace{\lambda(-2)}_{\neq 0} + \underbrace{\lambda(2)}_{\neq 0} + \dots \\ &= E[\varepsilon_t^2 x_t x_t'] + \sum_{h \neq 0} \lambda(h),\end{aligned}$$

so that

$$AVar(\hat{\beta}) = (E[x_t x_t'])^{-1} \left(E[\varepsilon_t^2 x_t x_t'] + \sum_{h \neq 0} \lambda(h) \right) (E[x_t x_t'])^{-1}.$$

Asymptotic Variance of $\hat{\beta}$: Estimators

- The expression of $AVar(\hat{\beta})$ varies depending on
 - Homoskedasticity or Heteroskedasticity
 - No serial correlation or Serial correlation
- Under Scenario 1 (homoskedasticity and no serial correlation), we use

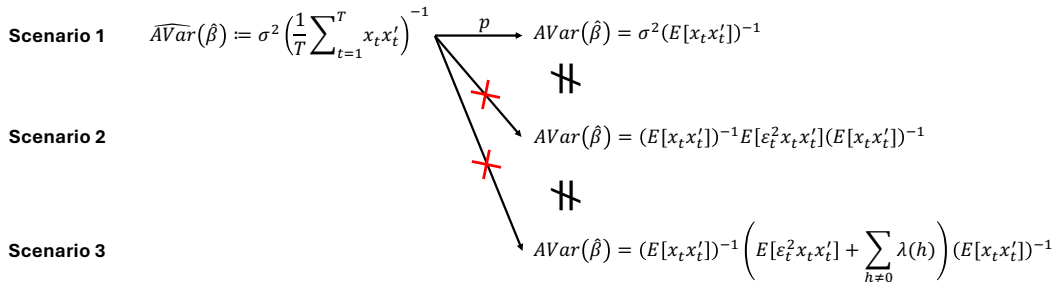
$$\widehat{AVar}(\hat{\beta}) := \hat{\sigma}^2 \left(\frac{1}{T} \sum_{t=1}^T x_t x_t' \right)^{-1}.$$

- In fact,

$$\widehat{AVar}(\hat{\beta}) \xrightarrow{P} AVar(\hat{\beta}).$$

- But, we cannot continue to use the same $\widehat{AVar}(\hat{\beta})$ in Scenarios 2 and 3.

Asymptotic Variance of $\hat{\beta}$: Estimators



Asymptotic Variance of $\hat{\beta}$: Estimators

Scenario 1 $\widehat{AVar}(\hat{\beta}) := \sigma^2 \left(\frac{1}{T} \sum_{t=1}^T x_t x_t' \right)^{-1} \xrightarrow{p} AVar(\hat{\beta}) = \sigma^2 (E[x_t x_t'])^{-1}$

$\nparallel$

Scenario 2 $\widehat{AVar}(\hat{\beta}) := ? \xrightarrow{p} AVar(\hat{\beta}) = (E[x_t x_t'])^{-1} E[\varepsilon_t^2 x_t x_t'] (E[x_t x_t'])^{-1}$

$\nparallel$

Scenario 3 $\widehat{AVar}(\hat{\beta}) := ? \xrightarrow{p} AVar(\hat{\beta}) = (E[x_t x_t'])^{-1} \left(E[\varepsilon_t^2 x_t x_t'] + \sum_{h \neq 0} \lambda(h) \right) (E[x_t x_t'])^{-1}$

Summary

	Homoskedasticity	No Serial Correlation	$\widehat{AVar}(\hat{\beta})$
Scenario 1	✓	✓	$\hat{\sigma}^2 \left(\frac{1}{T} \sum_{t=1}^T x_t x_t' \right)^{-1}$
Scenario 2	✗	✓	?
Scenario 3	✗	✗	?

- We study two approaches for these problems:
 1. Heteroskedasticity and Autocorrelation Consistent (HAC) Variance Estimator
 2. Generalized Least Squares (GLS) Estimation
- Insight: Scenario 3 $\supseteq$ Scenario 2 $\supseteq$ Scenario 1

Linear Regression of Time-Series Data III

**Heteroskedasticity and Autocorrelation Consistent (HAC)
Variance Estimator**

- Here are three insights:
 1. $\hat{\beta}$ is consistent for β under TS1'–TS3'.
 2. Homoskedasticity (TS4'-1) is a special case of heteroskedasticity.
 3. No serial correlation (TS4'-2) is a special case of serial correlation.
- Hence, we keep on using $\hat{\beta}$ with TS1'–TS3', while generalizing TS4'-1 and TS4'-2.

Setup

Assumption 4-1 (TS4''-1): Heteroskedasticity

$$E[\varepsilon_t^2 | x_t] = \sigma_t^2 \text{ for all } t \in \{1, \dots, T\}$$

Assumption 4-2 (TS4''-2): Serial Correlation

$$E[\varepsilon_t \varepsilon_s | x_t, x_s] \neq 0 \text{ for all } t, s \in \{1, \dots, T\}$$

- TS4''-1 and TS4''-2 are distinct:
 - TS4''-1: The variance of the errors varies over time or across different regimes.
 - TS4''-2: A shock has a lasting effect.

Proposition: Asymptotic Inefficiency

Under TS1'–TS3', TS4''-1, and TS4''-2, there exists a consistent and normally-distributed estimator $\hat{b}$ for β such that $AVar(\hat{\beta}) \geq AVar(\hat{b})$.

- $\hat{\beta}$ is no longer the most (asymptotically) precise estimator.

Asymptotic Variance of $\hat{\beta}$: Estimation (Roadmap)

- $AVar(\hat{\beta}) := (E[x_t x_t'])^{-1} \Omega (E[x_t x_t'])^{-1}$
- We know $(\frac{1}{T} \sum_{t=1}^T x_t x_t')^{-1} \xrightarrow{P} (E[x_t x_t'])^{-1}$.
- It remains to estimate Ω consistently.
- The following exposition proceeds in three steps:
 1. Develop a prototype estimator $\hat{\Omega}_M$ under Scenario 3
 2. Develop a more sophisticated estimator $\hat{\Omega}_{NW}$ under Scenario 3
 3. Application to Scenario 2 (heteroskedasticity and no serial correlation)

Asymptotic Variance of $\hat{\beta}$: Estimation ($\hat{\Omega}_M$)

- We want to estimate

$$\Omega = \sum_{h=-\infty}^{\infty} \lambda(h) \text{ with } \lambda(h) := E[\varepsilon_t \varepsilon_{t-h} x_t x'_{t-h}].$$

- **Problem:** $\sum_{h=-\infty}^{\infty}$ is not operational as h goes from $-\infty$ to ∞ .
- **Idea:** to truncate the sum at some point $\pm M$.
- We now target

$$\Omega_M := \sum_{h=-M}^M \lambda(h) \text{ with } \lambda(h) := E[\varepsilon_t \varepsilon_{t-h} x_t x'_{t-h}].$$

– M : the lag truncation number (a.k.a. the bandwidth)

- If M is “sufficiently large”, $\Omega \approx \Omega_M$ because $\lambda(h) \rightarrow 0$ as $h \rightarrow \infty$.

Asymptotic Variance of $\hat{\beta}$: Estimation ($\hat{\Omega}_M$)

- A natural estimator is the sample analog:

$$\hat{\Omega}_M := \sum_{h=-M}^M \hat{\lambda}(h),$$

where

$$\hat{\lambda}(h) := \frac{1}{T} \sum_{1 \leq t-h \leq T} \hat{\varepsilon}_t \hat{\varepsilon}_{t-h} x_t x'_{t-h}$$

with $\hat{\varepsilon}_t := y_t - x_t \hat{\beta}$.

Asymptotic Variance of $\hat{\beta}$: Estimation ($\hat{\Omega}_M$)

Proposition: Consistency of $\hat{\Omega}_M$

For any fixed M , $\hat{\Omega}_M \xrightarrow{P} \Omega_M$.

- Two practical issues about $\hat{\Omega}_M$.
 - (i) $\hat{\Omega}_M$ is a non-smooth function of M .
 - $\hookrightarrow \hat{\Omega}_M$ is sensitive to the choice of M ...
 - (ii) $\hat{\Omega}_M$ may not be positive semi-definite.
 - $\hookrightarrow \hat{\Omega}_M$ may not be a valid covariance matrix estimator...

Asymptotic Variance of $\hat{\beta}$: Estimation ($\hat{\Omega}_{NW}$)

- A more sophisticated estimator is a weighted sum of autocovariances:

$$\hat{\Omega}_{NW} := \sum_{h=-M}^M \left(1 - \frac{|h|}{M+1}\right) \hat{\lambda}(h).$$

- The weight function $(1 - \frac{|h|}{M+1})$ is called the Bartlett kernel.
- It is shown that
 - (i)' $\hat{\Omega}_{NW}$ is a smooth function of M .
 $\hookrightarrow \hat{\Omega}_M$ is not sensitive to the choice of M !
 - (ii)' $\hat{\Omega}_{NW}$ is positive semi-definite.
 $\hookrightarrow \hat{\Omega}_M$ is a valid covariance matrix estimator!

Asymptotic Variance of $\hat{\beta}$: Estimation ($\hat{\Omega}_{NW}$)

Proposition: Consistency of $\hat{\Omega}_{NW}$ (Newey and West, 1987)

$\hat{\Omega}_{NW} \xrightarrow{P} \Omega$ as long as M increases to infinity with T .

- Our variance estimator

$$\widehat{AVar}(\hat{\beta}) := \left(\frac{1}{T} \sum_{t=1}^T x_t x_t' \right)^{-1} \hat{\Omega}_{NW} \left(\frac{1}{T} \sum_{t=1}^T x_t x_t' \right)^{-1}.$$

- Then,

$$\widehat{AVar}(\hat{\beta}) \xrightarrow{P} AVar(\hat{\beta}).$$

- This is called
 - Heteroskedasticity and Autocorrelation Consistent (HAC) estimator
 - Heteroskedasticity and Autocorrelation Robust (HAR) estimator
 - Newey-West estimator (Newey and West, 1987)

Asymptotic Variance of $\hat{\beta}$: Estimation ($\hat{\Omega}_{NW}$)

- One last practical issue is the choice of the lag truncation number M .
- A rule of thumb:

$$M = \left\{ 6 \frac{\rho^2}{(1 - \rho^2)^2} \right\} T^{1/3},$$

where ρ is a serial correlation parameter.

- It is recommended to use $\hat{\rho}$ for ρ .
- An alternative is to use a default value of ρ :
 - if $\rho = 0.25$, then $M = 0.75 T^{1/3}$.
 - if $\rho = 0.50$, then $M = 1.40 T^{1/3}$.

Asymptotic Variance of $\hat{\beta}$: Application

- **Scenario 2:** Heteroskedasticity and No serial correlation

$$\Omega = \lambda(0) = E[\varepsilon_t^2 x_t x_t']$$

- By choosing $M = 0$,

$$\hat{\Omega}_{NW} = \hat{\lambda}(0) = \frac{1}{T} \sum_{t=1}^T \hat{\varepsilon}_t^2 x_t x_t',$$

so that

$$\widehat{AVar}(\hat{\beta}) = \left(\frac{1}{T} \sum_{t=1}^T x_t x_t' \right)^{-1} \left(\frac{1}{T} \sum_{t=1}^T \hat{\varepsilon}_t^2 x_t x_t' \right) \left(\frac{1}{T} \sum_{t=1}^T x_t x_t' \right)^{-1}.$$

- This is called
 - Heteroskedasticity Consistent (HC) estimator
 - Heteroskedasticity Robust (HR) estimator
 - Eicker–Huber–White estimator (Eicker, 1967; Huber, 1967; White, 1980)

Linear Regression of Time-Series Data III

Generalized Least Squares (GLS) Estimation

Quick Recap of GLS

- **Idea:** Transform the variables to satisfy the standard set of assumptions.
- Consider a time-series model:

$$y_t = x_t' \beta + \varepsilon_t,$$

where $\{\varepsilon_t\}$ possibly exhibit heteroskedasticity and serial correlation.

- For an arbitrary σ^2 , we can write $E[\varepsilon\varepsilon'|X] = \sigma^2 V(X)$.
- There always exists a $T \times T$ matrix $C(X)$ such that $C(X)'C(X) = V(X)^{-1}$.
- Consider the following transformation of the variables:

$$\tilde{y}_t = C(X)y_t, \quad \tilde{x}_t = C(X)x_t, \quad \tilde{\varepsilon}_t = C(X)\varepsilon_t.$$

- Now we have

$$\tilde{y}_t = \tilde{x}_t' \beta + \tilde{\varepsilon}_t.$$

GLS and Exogeneity Assumption

- Q. Are the standard set of assumptions satisfied for the transformed variables?

A. In general, no...

- Contemporaneous exogeneity is not preserved...
- Heteroskedasticity and serial correlation do not resolve...

Assumptions	Before Transformation	After Transformation	
		Ideal	Real
Linearity	✓	✓	✓
No Perfect Multicollinearity	✓	✓	✓
Contemporaneous exogeneity	✓	✓	✗
Homoskedasticity	✗	✓	✗
No Serial Correlation	✗	✓	✗

GLS and Exogeneity Assumption

- Consider a static model with an AR(1) error term:

$$y_t = \alpha + \beta x_t + \varepsilon_t,$$

with

$$\varepsilon_t = \rho \varepsilon_{t-1} + e_t,$$

where $e_t \sim WN(0, \sigma_e^2)$ and $|\rho| < 1$.

$$\implies E[\varepsilon_t^2] = \frac{1}{1-\rho^2} \sigma_e^2, \quad E[\varepsilon_t \varepsilon_{t-h}] = \frac{\rho^h}{1-\rho^2} \sigma_e^2$$

- Suppose contemporaneous exogeneity: $E[\varepsilon_t | x_t] = 0$ ($\implies E[\varepsilon_t x_t] = 0$)
- Q. Can we use GLS?

GLS and Exogeneity Assumption

- The variance and covariance can be arranged as

$$E[\varepsilon\varepsilon'] = \frac{\sigma_e^2}{1-\rho^2} \underbrace{\begin{bmatrix} 1 & \rho & \rho^2 & \dots & \rho^{T-1} \\ \rho & 1 & \rho & \dots & \rho^{T-2} \\ \vdots & \vdots & \vdots & \ddots & \rho \\ \rho^{T-1} & \rho^{T-2} & \dots & \rho & 1 \end{bmatrix}}_{=:V}.$$

- The matrix V^{-1} can be decomposed as $V^{-1} = C'C$ such that

$$C = \begin{bmatrix} (1-\rho^2)^{1/2} & 0 & 0 & \dots & 0 \\ -\rho & 1 & 0 & \dots & 0 \\ 0 & -\rho & 1 & \dots & \vdots \\ \vdots & \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \dots & -\rho & 1 \end{bmatrix}.$$

GLS and Exogeneity Assumption

- The transformation by C is called quasi-differencing: for $t \geq 2$,
 - $\tilde{y}_t = y_t - \rho y_{t-1}$
 - $\tilde{x}_t = x_t - \rho x_{t-1}$
 - $\tilde{\varepsilon}_t = \varepsilon_t - \rho \varepsilon_{t-1}$
- $\tilde{\varepsilon}_t$ may be correlated with $\tilde{x}_t$:

$$\begin{aligned} E[\tilde{\varepsilon}_t \tilde{x}_t] &= E[(\varepsilon_t - \rho \varepsilon_{t-1})(x_t - \rho x_{t-1})] \\ &= \underbrace{E[\varepsilon_t x_t]}_0 - \rho E[\varepsilon_t x_{t-1}] - \rho E[\varepsilon_{t-1} x_t] + \rho^2 \underbrace{E[\varepsilon_{t-1} x_{t-1}]}_0 \\ &\neq 0 \end{aligned}$$

- We have to replace contemporaneous exogeneity by strict exogeneity.
 - This rules out any model with lagged dependent variables.

GLS for Time-Series Data

- With strict exogeneity, the standard results of GLS hold for the time-series context.
- GLS estimator:

$$\hat{\beta}_{GLS} = (X' V X)^{-1} (X' V Y)$$

- unbiasedness, BLUE, **consistency**, **asymptotic efficiency**, asymptotic normality, etc...

- Feasible GLS estimator:

$$\hat{\beta}_{FGLS} = (X' \hat{V} X)^{-1} (X' \hat{V} Y)$$

- **consistency**, **asymptotic efficiency**, asymptotic normality, etc...

- In practice, $\hat{\beta}_{FGLS}$ and $\hat{V}$ are estimated in a recursive manner.
 - Estimating $\hat{V}$ requires a model for the error term.

GLS for Time-Series Data

Recursive Method of FGLS

For a static model with an AR(1) error term,

- (i) Regress y_t on $x_t \rightarrow$ Obtain $\hat{\varepsilon}_t$
- (ii) Regress $\hat{\varepsilon}_t$ on $\hat{\varepsilon}_{t-1} \rightarrow$ Obtain $\hat{\rho}$
- (iii) Calculate $\hat{\beta}_{FGLS}(\hat{\rho}) \rightarrow$ Obtain $\hat{\varepsilon}_t$
- (iv) Repeat (ii)–(iii) until $\hat{\beta}_{FGLS}(\hat{\rho})$ converges.

- Prais-Winsten estimator:
 - the first observation is included as suggested in the matrix C .
- Cochrane-Orcutt estimator:
 - the first observation is dropped.
 - loss of information

Linear Regression of Time-Series Data III

HAC vs. GLS

HAC vs. GLS

- HAC and (F)GLS can be compared from three perspectives.
- **Exogeneity:**
 - HAC does not rely on a specific type of exogeneity assumption.
 - GLS requires the strict exogeneity assumption.
 - ⇒ use HAC for any model with lagged dependent variables.
- **Efficiency:**
 - HAC does not improve the efficiency of $\hat{\beta}$.
 - GLS is asymptotically efficient.
 - ⇒ use GLS for asymptotically efficient estimation.
- **Knowledge about the error structure:**
 - HAC does not require knowledge about the error structure.
 - FGLS assumes a parametric structure on the error term.
 - ⇒ use HAC if the error structure is unknown.

HAC vs. GLS

- Ex1) Autoregressive Moving Average (ARMA) model of order p and q:

$$y_t = \beta_0 + \beta_1 y_{t-1} + \dots + \beta_p y_{t-p} + \varepsilon_t + \gamma_1 \varepsilon_{t-1} + \dots + \gamma_q \varepsilon_{t-q}$$

↪ HAC

- Ex2) Static model with an AR(p) error term:

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 z_t + \gamma_1 \varepsilon_{t-1} + \dots + \gamma_p \varepsilon_{t-p} + \varepsilon_t$$

↪ HAC or GLS

- Ex3) Static model with an unknown error structure:

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 z_t + g(\varepsilon_{t-1}, \dots, \varepsilon_{t-p}, \varepsilon_t)$$

↪ HAC

Linear Regression of Time-Series Data III

Two Caveats

Testing for Heteroskedasticity and Serial Correlation

- There are many statistical tests for heteroskedasticity and serial correlation:
 - Durbin-Watson test (Durbin and Watson, 1950)
 - Durbin's h statistics (Durbin, 1970)
 - Box-Pierce test (Box and Pierce, 1970)
 - Ljung-Box test (Ljung and Box, 1978)
 - Breusch-Godfrey test (Breusch, 1978; Godfrey, 1978)
- A common practice is to calculate the asymptotic variance under all possible specifications.

Application to Cross-Sectional Analysis

- Heteroskedasticity and serial correlation can also happen in the cross-sectional context:
 - Belgium, the Netherlands, France, and Germany
 - U.S., Canada, and Mexico
- Analogous analyses can be applied:
 - Spatial HAC
 - Cluster-Robust Standard Errors

Summary & Preview

Summary

- We extend the linear regression framework to accommodate models with a serially correlated error term.
- The expression of $AVar(\hat{\beta})$ changes depending on
 - Homoskedasticity vs. Heteroskedasticity
 - No serial correlation vs. Serial correlation
- We study two solutions:
 - Heteroskedasticity-Autocorrelation-Robust (HAC) Estimator
 - Generalized Least Squares (GLS) Estimation

Preview

- We have implicitly assumed that data are weakly stationary and weakly dependent.
- In many practical cases, these are not true...
- We will study
 1. tests of weak stationarity and weak dependence.
 2. linear regression analysis of non-stationary and non-weakly dependent processes.
- **Reminder:** Assignment 1 is due by 16:00, March 20 (Fri).
- **Next class:** March 27 (Fri), 16:00-17:15, Auditorium A - Jean-Norbert Cloquet. T2. Campus Ufo.

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Appendix

General Expression for the Asymptotic Variance of $\hat{\beta}$

- Let $v_t := x_t \varepsilon_t$.

$$\begin{aligned}\Omega &= \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \sum_{s=1}^T \underbrace{E[v_t v_s']}_{=: \lambda(t-s)} = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \sum_{s=1}^T \lambda(t-s) \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{h=-(T-1)}^{T-1} (T - |h|) \lambda(h) = \lim_{T \rightarrow \infty} \sum_{h=-(T-1)}^{T-1} \left(1 - \frac{|h|}{T}\right) \lambda(h) \\ &= \sum_{h=-\infty}^{\infty} \lambda(h),\end{aligned}$$

where $\lambda(h) := E[v_t v_{t-h}']$.